

Kai Teng Hou (Harry)

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EDUCATION

Pennsylvania State University

B.S. in Computer Science, Minor in Mathematics, Nittany AI Society

University Park, PA

Expected May 2027

EXPERIENCE

AI Generative Technology Research Intern

Yahoo!

July 2025 – Oct 2025

Typescript, Ruby, GraphQL

- Rotated across **Sales, Marketing, Operations**, and **Finance** departments to identify workflow bottlenecks, providing actionable recommendations that optimized processes through automation and **AI-driven solutions**.
- Designed and implemented a **real-time price tracking system** for sales assistants, leveraging **web scraping** and **AI-powered product matching** to monitor competitor pricing instantly, **reducing manual research time by 70%** and enabling **same-day pricing adjustments**.
- Collaborated with cross-functional teams to integrate AI-assisted tools, streamlining data handling, reducing manual workload, and improving decision-making efficiency.

Machine Learning Research Assistant

Penn State University

February 2025 – Present

Python, Computer Vision, Data Processing

- Collaborated with Dr. Enrico Casella on a smart agriculture project, annotating and preprocessing over 500 images using **Roboflow**, enabling effective training of **YOLOv8-seg** models for turkey segmentation.
- Optimized segmentation accuracy (+20%) and YOLOv8 mAP50-95(M) (+15%) by refining datasets and automating model evaluation pipelines using **Python**, hyperparameter tuning, and **IoU**-based performance metrics.

Software Engineer Intern

PenguinSmart (Start-up)

July 2024 – Nov 2024

Typescript, Ruby, GraphQL

- Designed and developed web applications to enhance user experience and functionality, utilizing **React**, **Node.js**, and **Ruby on Rails** based on existing designs.
- Revamped admin UI by implementing data filtering, sorting, and video upload functionality, improving operational efficiency and **reducing data retrieval time by 30%** and improving workflow efficiency for admins.
- Delivered core features, including a name-editing functionality and enhanced **Google Authentication** with improved error handling and redirects.
- Coordinated with cross-functional teams and utilized **Bitbucket** using **Git** for version control and **JIRA** for issue tracking.

PROJECTS

Tutorific.AI | *React, JavaScript, OpenAI API* | github.com/TutorificAI

- Partnered with a teammate to build an educational assistant platform that offers personalized tutoring, flashcards, writing assistance, and customizable quizzes, achieving **3rd place at HackPSU Fall 2024 out of 69 teams**.
- Co-designed an intuitive UI with smooth transitions and visually engaging animations, incorporating distinct icons and colors for each feature.
- Integrated Q&A support, feedback, and flashcard review functionalities to foster active learning and improve concept retention.
- Enabled topic-specific quizzes, including multiple-choice, true/false, and short-answer formats with instant feedback and scoring capabilities.

TECHNICAL SKILLS

Languages: Python, Java, Typescript, Ruby, C, Swift, JavaScript/HTML/CSS

Frameworks (Libraries): React, Node.js, GraphQL, Selenium, Appium, Ruby on rails

Tools : Git, JIRA